SALT Cap Workarounds*

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October 2025

Abstract

The 2017 Tax Cuts and Jobs Act considerably limited federal deductions for state and local tax (SALT) payments, generally to $10,000 for individual taxpayers. In response, many states enacted passthrough entity tax (PTET) laws to facilitate a workaround to this limitation. PTET laws, in effect, allow passthrough business owners to remit their state individual income taxes via their businesses, where SALT deductions are not subject to any limit. Using this workaround, we estimate that passthrough businesses deducted $70 billion of income taxes in 2024, translating to $18 billion in reduced federal revenue and largely benefiting the very top of the income distribution.

Key Words: Tax Policy, Tax Cuts and Jobs Act, One Big Beautiful Bill Act, SALT Deduction, Passthrough Business

JEL Codes: H24, H25, H26, H68, H71, H77

*We thank James Cilke for invaluable help in using the Joint Committee on Taxation's Individual Tax Model to obtain several estimation parameters, and thank Edgar Aguirre for excellent research assistance. We also thank Thomas Barthold, James Cilke, Stephen Coate, John Iselin, Miles Johnson, Greg Leiserson, Jeremy Lent, Jake Mortenson, James Pearce, Molly Saunders-Scott, Thalia Spinrad, and David Splinter for helpful comments. This research embodies work undertaken by the staff of the Joint Committee on Taxation, but as members of both parties and both houses of Congress comprise the Joint Committee on Taxation, this work should not be construed to represent the position of any member of the Committee. Email addresses: matthew.comey@jct.gov (Comey); brian.gallagher@jct.gov (Gallagher); william.gorman@jct.gov (Gorman).

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1 Introduction

The U.S. federal income tax system interacts heavily with individual state tax policies, in large part through the ability for taxpayers to deduct state and local tax (SALT) payments when computing federal taxable income (Feenberg and Rosen, 1986; Feldstein and Metcalf, 1987). In a monumental development on this front, Public Law 115-97, commonly known as the 2017 Tax Cuts and Jobs Act (TCJA), significantly reduced the extent to which individuals could deduct SALT payments. First, TCJA roughly doubled the standard deduction, leading many taxpayers to forego itemizing deductions entirely and thereby making the SALT deduction unavailable. Second, TCJA instituted a "SALT cap," limiting most individual SALT deductions to $10,000 per return ($5,000 for married individuals filing separately). Largely as a result of the increased standard deduction and the SALT cap, aggregate individual SALT deductions fell from $625 billion in 2017 to just $146 billion in 2018 (Statistics of Income, 2019, 2020). The SALT cap was somewhat loosened with the recent passage of Public Law 119-21, commonly known as the 2025 One Big Beautiful Bill Act (OBBBA). OBBBA temporarily increases the cap to $40,000 ($20,000 for married individuals filing separately), with a phasedown to $10,000 for higher income taxpayers, but even this modified cap is a substantial constraint relative to pre-2018 policy.

Immediately upon passage of TCJA, states began to look for ways to reduce their residents' federal tax burdens via SALT cap workaround laws, which generally aim to reclassify state individual income tax payments as items that are more broadly deductible at the federal level. By far the most successful of these attempts has been the passthrough entity tax (PTET) regime, though we discuss other workaround laws in Section 2, below.[1] In effect, PTET regimes allow owners of passthrough businesses — S corporations and partnerships — to remit state individual income taxes via their businesses. Generally, passthrough businesses are not subject to entity-level state or federal income tax; instead, owners pay individual income tax on shares of net income that are "passed through." This arrangement contrasts with the tax treatment of C corporations, which are directly subject to corporate income taxes. Under a PTET regime, however, a passthrough business may elect to pay a business income tax in exchange for tax benefits afforded to its owners, either in the form of tax credits or income exclusions on their individual state income tax returns. PTET laws are constructed to be approximately revenue-neutral for the state, with PTET liabilities corresponding to

[1] We use "PTET" to specifically refer to a workaround tax paid by passthrough entities, but other non-workaround entity taxes levied on passthrough businesses do exist, primarily in the form of franchise taxes.

the reduction in owners' individual income tax liabilities. However, PTETs reduce federal revenues, since passthrough businesses may deduct PTETs without limitation on federal business tax returns. By deducting these taxes at the entity level, taxable passthrough income that flows to owners is reduced by the full value of the PTET, thereby allowing an uncapped SALT deduction for what is, for all practical purposes, an individual income tax. By 2024, 36 states had adopted a PTET regime. In this paper, we quantify the magnitude of PTET deductions, estimate the corresponding fiscal cost to the federal government, and allocate the individual tax benefit of these workarounds across the income distribution.

It is difficult to evaluate the extent to which PTETs are used to avoid the SALT cap. Published state-level data on PTET receipts are often out of date or, in many cases, completely unavailable. At the federal level, a PTET deduction is invisible on an individual tax return; the workaround operates via a reduced amount of passthrough income reported on individual tax returns, since the deduction is claimed by businesses before income is allocated to owners. On federal business tax returns, PTETs are not explicitly reported as a deduction item, but are rather added into the broader "Taxes & Licenses" deduction line. To learn about these deductions, we use a representative sample of business tax returns from the IRS Statistics of Income Division (SOI), described in Section 3, below. We focus on the "Taxes & Licenses" deduction line, but since this line aggregates a number of payments made by businesses to government entities, we employ a difference-in-differences research design to assess how the deduction evolved after the enactment of PTET regimes. Specifically, we compare aggregate taxes and licenses deductions claimed by passthrough businesses in states with and without PTETs, before and after enactment of the laws. Under a parallel trends assumption, this design will return the effect of PTET regimes on passthrough business tax deductions. Our preferred specification is a triple difference design (difference-in-difference-in-differences), which further differences out trends in taxes and licenses deductions taken by C corporations, which cannot elect to pay PTETs. This allows us to control for unobservable factors that may affect the deduction differentially across states and years. We discuss this strategy in more detail in Section 4.

We then use our estimates to construct an implied fiscal cost of these SALT cap workarounds, detailed in Section 5. A fiscal cost emerges because, in the absence of the workarounds, federally taxable business income would not be reduced by the value of PTETs and business owners would only be able to deduct corresponding state individual income taxes to the extent they are not limited by the SALT cap. To estimate this cost, we first convert our causal estimates into implied amounts of PTET deductions included on federal business tax

returns. Then, we simulate the tax consequences of denying these deductions to passthrough businesses and allocating them back to individual taxpayers. Overall, we estimate that in 2024, there were $69.8 billion in federal passthrough business deductions related to PTETs, $64.0 billion of which corresponded to deductions that would otherwise have been classified as individual SALT. Those diverted deductions would have been almost entirely disallowed on individual tax returns and amount to 29 percent of SALT deductions that were actually disallowed due to the cap. This translates to $18.1 billion in foregone federal revenue for tax year 2024. In addition to being costly, the PTET workarounds are only available to owners of passthrough businesses, who are disproportionately located at the very top of the income distribution. We estimate that about 80 percent of PTET benefits flow to taxpayers in the top 1 percent.

SALT cap workaround laws have generated substantial interest among tax policy experts and federal lawmakers. Even without the causal estimates presented in this paper, it has been apparent to experts that these workaround laws are costly to the federal government (e.g., Rueben, 2021). Other commentators, such as the Wall Street Journal Editorial Board, have noted that PTETs are distortionary, providing a large tax benefit to owners of passthrough entities but no comparable benefit to owners sole proprietorships or C corporations.[2] Advocates for owners of passthrough businesses, however, argue that PTET deductibility is desirable for maintaining parity with C corporations, which can deduct state income taxes without limit.[3] However, unlike PTETs, corporate income taxes do not generate corresponding state tax benefits for individual shareholders. In 2025, Congress moved toward limiting SALT cap workarounds via provisions included in early drafts of OBBBA, which we discuss in more detail in Section 6. While the bill that was initially passed by the House of Representatives included an anti-workarounds provision, the Senate ultimately dropped the provision and the workarounds remain permissible.

Our paper is related to several recent economics papers that document business responses to provisions of TCJA (e.g., Chodorow-Reich et al., 2024; Kennedy et al., 2024; Gale and Haldeman, 2021) as well as market responses to the SALT cap specifically (e.g., Kessler and Bruce, 2024; Horton, 2023). We add to this literature by showing how the individual SALT cap had spillover effects on businesses via state government responses. Our paper also relates to a growing literature documenting features of passthrough businesses more broadly, which had been little-studied in economics until recently (e.g., Smith et al., 2019, 2022). We

[2]See "Closing the SALT Business Loophole," published by the Wall Street Journal on June 24, 2025.

[3]See, for example, "WSJ is Wrong on SALT Parity," from the S Corp Association, a lobbying group representing S corporation shareholders, published on June 25, 2025.

highlight an additional tax advantage for structuring a business as a passthrough entity and underscore government's willingness to accommodate these business owners through legislative and regulatory action. At the intersection of these literatures is Goodman et al. (2025), who study passthrough business responses to the TCJA-instituted individual deduction for qualified passthrough business income. They generally find that businesses respond on low-cost margins, such as income realization and characterization, but find limited responses on high-cost margins related to real economic activity. Our paper highlights an extreme example of a low-cost behavioral response to changes in passthrough income taxation: electing to pay a PTET. Finally, our paper adds to the long list of examples of interplay play between U.S. state and federal income tax systems, a strand of literature exemplified by Feldstein and Metcalf (1987).

2 The SALT Deduction, Limitations, and Workarounds

The SALT deduction is as old as the U.S. federal income tax itself, with roots that trace back to the Union's efforts to fund the Civil War via the Revenue Acts of 1861, 1862, and 1864.[4] While the statutes stated that the deduction was applicable to taxes on income-producing property, the Revenue Acts were later deemed to provide a deduction for SALT assessed on both business and nonbusiness income. The Office of the Commissioner of Internal Revenue, a predecessor to the IRS, issued a ruling that allowed all SALT as a deduction, and Congress later codified the ruling in the Revenue Act of 1865 (Boutwell, 1864). While Congress repealed the income tax in 1872, the SALT deduction reemerged as a permanent part of the U.S. federal income tax system in the Tariff Act of 1913.

Since 1913, Congress has curtailed the SALT deduction to exclude or otherwise limit deductions for nonbusiness SALT. For example, Congress enacted the standard deduction as part of the Individual Income Tax Act of 1944, eliminating the nonbusiness SALT deduction for most individuals due to their standard deduction exceeding itemized deductions.[5] For taxpayers who do itemize, Congress has limited the SALT deduction in other ways, either directly or indirectly. For instance, it removed nonbusiness SALT deductions for auto license taxes and certain excise taxes in 1964, and for gasoline and motor fuel taxes in 1978.[6] In the Tax Reform Act of 1986, Congress modified the alternative minimum tax (AMT) regime

[4]For example, the Revenue Act of 1861 provided "That, in estimating said income, all national, state, or local taxes assessed upon the property, from which the income is derived, shall be first deducted."

[5]For 2022, 88.5 percent of all tax returns claimed a standard deduction (Statistics of Income, 2025a).

[6]These provisions are included in the Revenue Act of 1964 and the Revenue Act of 1978, respectively.

to disallow taxpayers from deducting nonbusiness SALT from alternative minimum taxable income. Congress has also enacted limitations on the overall value of itemized deduction, which indirectly limits the value of SALT deduction.[7]

The most prominent limitation on SALT deductions was the TCJA $10,000 cap ($5,000 for married individuals filing separately) on certain real property, personal property, income, and sales taxes.[8] The $10,000 cap was applicable to both single filers and married couples who file jointly, and the dollar value was not indexed to inflation. Beginning in 2025, OBBBA temporarily raises the SALT cap to $40,000 ($20,000 for married individuals filing separately) for taxpayers with modified adjusted gross income less than $500,000, but the cap phases down to $10,000 between $500,000 and $600,000.[9] The SALT cap is scheduled to completely revert to $10,000 beginning in 2030. The $10,000 SALT cap was arguably the most restrictive SALT policy enacted to date. For example, the Joint Committee on Taxation (JCT) estimated that if the SALT cap were to have expired after 2025 (as originally scheduled under TCJA), tax expenditures related to individual SALT deductions would have risen from $22.6 billion in fiscal year 2025 to $197.1 billion in fiscal year 2027 (Joint Committee on Taxation, 2024).

Soon after passage of TCJA, high-tax states whose residents were disproportionately impacted by the SALT cap responded forcefully. Connecticut, Maryland, New Jersey, and New York filed a lawsuit against the SALT cap on constitutional grounds, arguing that the law violated the Tenth Amendment, which places restrictions on Congress's ability to force a state to legislate or expend resources against its will, and the Sixteenth Amendment, which governs Congress's power to tax.[10] The U.S. District Court for the Southern District of New York ruled for the federal government, holding that the SALT cap was within Congress's broad power to tax and did not coercively force states to change their tax policy. The Second Circuit upheld the decision[11] and the Supreme Court ultimately declined to review it.

As an early attempt at a SALT cap workaround, New York, New Jersey, and Connecticut enacted laws allowing taxpayers to make charitable contributions in exchange for state or lo-

[7]These actions include the now-expired "Pease" limitation (named after former Congressman Donald Pease), enacted as part of the Omnibus Budget Reconciliation Act of 1990, and the 2025 OBBBA provision to limit the tax benefit of itemized deductions to 35 percent of the value of deductions.

[8]Under TCJA, an individual may claim an itemized deduction up to the cap for the aggregate of (1) nonbusiness real and personal property taxes and (2) State and local income taxes (or sales taxes in lieu of income taxes). For taxes attributable to a business, there is no individual-level limitation on State and local property taxes, sales taxes, and other non-income taxes.

[9]Both the SALT cap and modified gross income amounts are increased by 1 percent annually until 2029.

[10]*New York v. Mnuchin*, 408 F. Supp. 3d 399 (S.D.N.Y. 2019).

[11]*New York v. Yellen*, 15 F.4th 569 (2d Cir. 2021).

cal tax credits. The goal of these laws was to allow individuals to shift their state individual income tax liability toward charitable contributions, which generally face a less restrictive deduction limitation of up to 60 percent of adjusted gross income. For example, New York enacted a law in 2018 that allowed donors to claim a credit for 85 percent of qualified contributions. In response, the U.S. Treasury Department issued regulations in 2019 requiring a taxpayer to reduce any charitable contribution by the amount of the SALT credit in connection with the donation. An initial challenge to those regulations was unsuccessful,[12] but an appeal is still pending.

New York also instituted a payroll tax workaround in 2018, via the Employer Compensation Expense Program (ECEP). The ECEP allows employers to pay an additional payroll tax on certain electing employees' wages, and those electing employees would then receive a state tax credit equal to the value of the payroll tax paid by the employer. Employers would then reduce wages to cover the cost of the elective tax. The system effectively allows workers to "deduct" state income taxes by way of lower wages. The ECEP, however, has proved unpopular: New York reported just $5 million in ECEP wage credits in fiscal year 2022 compared to $13 billion in PTET credits (New York State Division of the Budget, 2025). Low takeup may be attributed to worker reluctance to accept a wage reduction and the fact that the ECEP does not cover self-employed individuals, a group particularly impacted by the SALT cap. The Treasury Department has not provided guidance on the payroll tax workaround to date, likely in part due to its limited use.

By far, PTET regimes have proved to be the most successful of the SALT cap workaround attempts. Initially enacted in 2018 by Connecticut, 35 other states and New York City have adopted a PTET as of 2025. Under a PTET regime, a passthrough entity pays a state income tax, while owners (partners of partnerships or shareholders of S corporations) receive a corresponding individual income tax benefit proportionate to their share of the entity's PTET. Passthrough entities then deduct the full value of PTETs at the federal level, effectively allowing owners to deduct their state individual income tax liability from their federal ordinary business income. PTETs can operate either as a credit regime or an income exclusion regime. Under a credit regime, a state assesses an income tax on a passthrough business's income, but awards a corresponding full or partial credit to the individual owners.[13] We present an example of a hypothetical PTET benefit under a credit regime in Table 2. Column 1 shows the computation of state and federal tax under no PTET

[12] *New Jersey v. Mnuchin*, 133 AFTR 2d 2024-1240 (S.D.N.Y. 2024).

[13] As an example of a partial credit, Massachusetts only allows a credit equal to 90% of an owner's PTET share, allowing it to capture some revenue from the PTET.

regime, while Column 2 shows the computation under the PTET regime. Here, consider an S corporation ("Passthrough") with a single shareholder ("Shareholder"), operating in State A, which has a flat 5 percent individual income tax rate, a 5 percent PTET rate, and a 100 percent PTET credit. Assume that Passthrough earns $1 million of pre-tax business income. For State A income tax purposes, Passthrough pays a $50,000 PTET and allocates $1 million (the full amount of pre-tax business income) to Shareholder, such that he owes a tentative state income tax of $50,000. However, that tax is then reduced by the $50,000 PTET credit that Shareholder is entitled to. For State A, this is equivalent to taxing Shareholder's passthrough income at 5 percent in the absence of a PTET regime. For federal income tax purposes, Passthrough deducts the $50,000 PTET on its S corporation tax return (Form 1120-S) and allocates $950,000 ($1 million less $50,000 PTET) to Shareholder. This, in effect, allows Shareholder to deduct his full state income tax liability, rather than only $10,000 under the no-PTET counterfactual.[14] Assuming a federal average tax rate of 35%, this corresponds to $14,000 in tax savings.

Alternatively, under an income exclusion PTET regime, a state allows passthrough business owners to exclude their distributive share of business income on their individual state income tax returns (i.e., income on which the PTET was assessed at the entity level).[15] Modifying the above example, assume that rather than awarding a PTET credit, State B allows Shareholder to exclude their entire business income that is allocated from Passthrough. Here, Shareholder excludes the full $1 million from his State B taxable income; thus, there is no tentative tax. The net result is the same as the credit scenario: State B collects $50,000 and Shareholder receives federal income tax savings of $14,000 via the deductibility of the PTET from federal passthrough business income.

We summarize the proliferation and structure of PTET laws in Table 1, where we also present each state's top marginal tax rate (MTR) on individual passthrough business income. The table shows that most (but not all) states with an individual income tax have enacted a PTET as of 2025, and that the PTET rates generally (but not always) match the state's top marginal income tax rate. The four states with the highest MTRs have enacted a PTET, including California (13.3 % top MTR) and New York (10.9% top MTR). Delaware, Maine, North Dakota, Pennsylvania, Vermont, and Washington, DC have income taxes but have not enacted a PTET law.

[14]PTET elections are also advantageous to taxpayers who elect to take the standard deduction, as it allows for a SALT deduction when it otherwise would have been unavailable.

[15]Under exclusion regimes, states can also capture revenue from the PTET, as illustrated by Wisconsin, which sets it PTET rate slightly higher than its top marginal tax rate.

Initially, taxpayers were uncertain about the validity of PTETs as a workaround to the SALT cap, as it depended on a taxpayer-favorable interpretation by the IRS on the concepts of "separately stated" and "non-separately stated" passthrough business income. While passthrough businesses aggregate ordinary business income and allocate it to owners as a lump sum (i.e., it is non-separately stated), passthrough businesses must disaggregate and separately state certain income and deduction items because they are potentially subject to differential tax treatments for owners. For example, some income items may be subject to different tax rates (e.g., qualified dividends and long-term capital gains), while some deduction items face individual-level limitations (e.g., limits on charitable contributions). Some items, such as short-term and long-term capital gains are expressly identified as separately stated items in statute, whereas others are governed by regulation, including regulations that provide a "catch-all" provision requiring partnerships to separately state any item that "would result in an income tax liability for [a] partner, or for any other person, different from that which would result if that partner did not take the item into account separately."[16] A PTET that is characterized as a separately stated item is of little use to an individual: the recipient owner would be forced to report the separately stated PTET as an itemized individual SALT deduction, potentially subject to the SALT cap. The Treasury and IRS resolved this uncertainty by issuing informal guidance accepting the treatment of PTETs as a non-separately stated item in Notice 2020-75. The Notice cites an old revenue ruling in support of this position, which held that Cincinnati taxes imposed upon and paid by a partnership on business income are deductible from the partnership's non-separately stated taxable income.[17]

3 Data

To conduct our empirical analysis of PTET regimes, we use IRS business tax return data. Specifically, we combine the SOI Corporate and Partnership Samples to construct a representative sample of S corporations, partnerships, and C corporations for tax years 2014-2022.[18] Selection into the SOI samples is based on stratified random samples of income tax returns or other forms filed, with certain classes of returns sampled at higher rates as to allow the study of tax issues affecting small populations. The SOI samples are commonly used in tax

[16] Treas. Reg. sec. 1.702-1(a)(8)(ii).

[17] Rev. Rul. 58-25, 1958-1 C.B. 95

[18] Additionally, we use the Individual Sample to examine sole proprietorships in a robustness check, but defer details of this sample to Appendix A.

policy analysis and economics research to produce analyses that, if properly weighted, are representative of the population of U.S. taxpayers. For more information on SOI's sampling methodology, see Statistics of Income (2025b).

The SOI Corporate Sample is composed of entities which file corporate income tax returns via the various versions of Form 1120. This includes C corporations, S corporations, life insurance companies, and real estate investment trusts, among a few other less common entity types. S corporations are the only type of passthrough business entity represented in the Corporate Sample. The Partnership Sample is made up of entities filing partnership income tax returns via Form 1065. In our main analysis, we append together the samples of C corporations, S corporations, and partnerships in our nine tax years, totaling approximately 2.1 million business tax returns, or about 240,000 per year. These returns represent approximately 183 million total business returns, or about 20.3 million per year.

Because the SOI samples do not constitute a balanced panel of returns, we aggregate return information as to facilitate the use of panel regression techniques.[19] Specifically, since PTET laws are passed by individual states at different points during this time period and apply only to passthrough entities, we aggregate tax return information into state-year-entity type cells.[20] For our primary analysis, we focus on the aggregate value of the "Taxes & Licenses" deduction line within each state-year-entity type cell, but collect other income and deduction information for auxiliary analyses. We then adjust the dollar amounts for inflation using the Federal Reserve's Personal Consumption Expenditures price index. We defer additional details on processing of the SOI data to Appendix A.

We then supplement the SOI business return data with PTET-related policy information, including whether the state enacted a PTET during our sample period, the effective year of the PTET, and the enactment date.[21] We collect this information for all 50 states and the District of Columbia, which we summarize in Table 1 and Appendix Table B.1. As suggestive evidence of the importance of PTETs for federal taxation, Figure 1 shows aggregated passthrough entity taxes and licenses deductions, adjusted for inflation and relative to 2017, stratified by whether the state had enacted a PTET law by 2022 (the end of our sample period). The figure shows that from 2017 to 2022, aggregate deductions for taxes and licenses

[19] While not explicitly a panel, due to the stratified sampling methodology, many businesses appear every year.

[20] Specifically, we aggregate by calendar year, which in most instances aligns with a tax year. In instances where a business elects for a tax year ending on June 30 or earlier, we assign the observation the year preceding the year in which its tax year ends. Otherwise, we assign to the calendar year corresponding to the end of the tax year.

[21] In some cases, PTET policies were made retroactively effective, as summarized in Appendix Table B.1.

increased by 36% for passthrough entities in states with a PTET regime, compared to just a 17% in states without a PTET regime.

4 Causal Effects of PTET Regimes

We begin our empirical analysis by estimating the effect of state PTET regimes on deductions claimed by passthrough businesses. Specifically, passthrough entities may deduct state and local income taxes on the "Taxes & Licenses" line of business income tax returns. Because this line could contain any number of payments to state and local governments, we employ a difference-in-differences analysis to evaluate how the size of this deduction grew following states' passage of PTET laws. Our preferred specification is a triple difference design, where we construct counterfactual deductions using trends for both passthrough businesses in non-PTET states and C corporations in PTET states (which cannot benefit from a PTET). We use a triple difference design rather than a standard difference-in-differences design due to an observed upward trend in taxes and licenses deductions across states and entity types during our sample period, including deductions from C corporations in both PTET and non-PTET states, as illustrated in Appendix Figure C.1. Hence, further differencing out C corporation deductions within states should better control for trends in taxes and licenses deductions that are unrelated to the passage of PTET laws.[22]

To implement this design, we estimate a regression of the following form:

$$\ln(Y_{ste}) = \alpha + \beta PTET_{ste} + \gamma_{st} + \phi_{se} + \xi_{te} + \varepsilon_{ste}, \tag{1}$$

where s indexes states, t indexes years, and e indexes entity types (here, partnerships, S corporations, and C corporations). Our outcome variable of interest, $\ln(Y_{ste})$, is the log of taxes and licenses deductions, aggregated by state, year, and entity type. The independent variable of interest, $PTET_{ste}$, is an indicator for whether a PTET law is in effect for a given state-year-entity type observation and always takes a value of zero for entities in states that have not passed a PTET as of 2022, as well as for all C corporation observations.[23]

[22]Some states changed corporate income tax rates in the aftermath of TCJA, but we have no reason to suspect that the implementation or timing was correlated with the passage PTET laws. Nonetheless, we test robustness to estimating the triple difference with respect to sole proprietorships and obtain very similar results.

[23]Since our SOI data does not effectively capture amended returns, we define policies as starting in the later of the effective year and the enactment year.

Finally, the regression includes three sets of fixed effects to facilitate the triple difference research design: γ_{st} are state-by-year fixed effects, to control for within-state deduction trends common to all entity types; ϕ_{se} are state-by-entity type fixed effects, to control for within-state level differences in deductions for each entity type; and ξ_{te} are year-by-entity type fixed effects, to control for across-state trends in deductions for each entity type. All regressions are weighted using year 2017 ordinary business income, aggregated to state-by-entity type cells, which gives greater weight to PTET states with more pre-PTET business activity.[24] All standard errors are clustered by state.

Because PTET laws have been enacted over a span of several years, this research design can be classified as a "staggered-timing" difference-in-differences design. Recent econometrics research has demonstrated pitfalls of using multi-dimensional fixed effects regressions, like the specification shown in Equation 1, to estimate causal effects with such designs (e.g., Goodman-Bacon, 2021; Baker et al., 2022). The problem stems from earlier-treated units being implicitly used to construct counterfactual outcomes for later-treated units, which can pose a problem if treatment effects are not constant over time. To correct for this, we estimate treatment effects using a "stacked" regression in the style of Cengiz et al. (2019).[25] To implement the stacked regression, we first construct separate "datasets," where each treated state is paired with the complete set of never-treated control states.[26] We then append the datasets together and estimate a pooled treatment effect (β) from a single regression.

When constructing the datasets, we define the "never-treated" states as those that never implemented a PTET law but also those that did not implement such a law by 2022 (the end of our sample period), which, upon inspection of Table 1, includes a set of states diverse on a number of political, demographic, and tax policy dimensions. Each dataset includes four pre-treatment years and two post-treatment years, and we drop treated states that did not enact a PTET law by 2021, as to ensure two years of post-treatment data. We also drop Alaska and Hawaii, as is common in the public finance literature. Finally, in our main specification, we drop states with a PTET law operating through an income exclusion as opposed to a credit, as to ensure treatment status is relatively uniform. Our results are highly robust to this choice, however. This results in 13 treated states (and, correspondingly, 13

[24] Our results are nearly identical when we weight by aggregate SOI sampling weights in each cell, which corresponds to weighting each state by number of business entities (within entity types).

[25] See Deshpande and Li (2019) and Butters et al. (2022) for other recent examples.

[26] Wing et al. (2024) discuss various implementations where stacked difference-in-differences designs do not estimate an average treatment effect with sensible properties. They note that the multi-dimensional fixed effects regression framework, employed in our analysis, is suitable in cases where the share of treated units is constant across datasets, which we accomplish by pairing the same set of control states with each treated state.

datasets) in our primary specification and 17 treated states when adding back in the PTET exclusion states.

For β in Equation 1 to correspond to the causal effect of interest, we rely on a parallel trends assumption that may be framed as follows: after differencing out within-state trends in taxes and licenses deductions claimed by C corporations, we require that, in the absence of PTET laws, the deductions claimed by passthrough entities would have trended identically in PTET and non-PTET states.[27] The inclusion of C corporations in our research design is quite useful for constructing counterfactual growth in taxes and licenses deductions, since these businesses face many of the same types of non-PTET tax and license expenses as the passthrough businesses, including real estate taxes, payroll taxes, and sales and excise taxes on the purchase of intermediate goods, all of which vary across years and states.

While the parallel trends assumption is fundamentally untestable, coefficients from an event study regression can demonstrate whether entities with and without exposure to a PTET regime were trending similarly in the years leading up to the passage of the PTET regime. Parallel pre-trends are suggestive of trends that would have remained parallel in the absence of a policy change. To demonstrate this, we estimate a regression of the following form:

$$\ln(Y_{ste}) = \alpha + \sum_{\tau=-4}^{\tau=1} \beta^{\tau} PTET_{ste} \times \mathbb{1}(t = \tau) + \gamma_{st} + \phi_{se} + \xi_{te} + \varepsilon_{ste}, \tag{2}$$

where we estimate separate β^{τ} coefficients for periods spanning four years prior to PTET passage ($\tau = -4$) to two years following passage ($\tau = 1$), omitting estimation of β^{-1}, which sets the reference year for relative effects to one year before passage.

Difference-in-differences research designs also rely on a stable unit treatment value assumption (SUTVA), which requires that the composition of treated and control units is fixed over time. Two possible SUTVA concerns in our setting are business relocation and business entity-type switching. First, a SUTVA violation would occur if business relocation shifted passthrough income (and deductions) between PTET and non-PTET states in a manner that was correlated with passage of these laws. We believe this is on its face unlikely, as a PTET is simply an undoing of a tax shock created by the SALT cap, resulting in a relatively small change in net tax liability once the PTET is enacted. Further, in studying the TCJA Section 199A deduction for certain passthrough income, Goodman et al. (2025) generally

[27] Equivalently, we may frame the parallel trends assumption as: after differencing out across-state trends in taxes and licenses deductions paid by all entity types, we require that, in the absence of PTET laws, the deductions claimed by passthrough entities and C corporations would have trended identically.

find that passthrough businesses exhibit limited behavioral response to tax policy changes on margins that are costly to change, as would be the case with relocation. Additionally, Kessler and Bruce (2024) find no effect of the SALT cap on interstate migration. A second possible SUTVA violation is related to entity-type switching, where non-passthrough entities reclassify as passthrough entities, or where new businesses disproportionately classify as passthrough entities. We provide evidence that this was not the case in Appendix Figure C.2, where we show that within PTET states, aggregate entity gross receipts trended almost identically for passthrough entities and C corporations after 2017. Further bolstering this claim, Kennedy et al. (2024) find little evidence of entity-type switching in the aftermath of TCJA, which substantially changed the tax incentives surrounding entity type classification.

Moving to results, our triple difference and event study regression coefficients are summarized in Figure 2. We estimate the effect for S corporations (top panel) separately from partnerships (bottom panel), motivated by the observation that base-level taxes and licenses deductions are larger for S corporations than partnerships and, hence, percent changes may be difficult to interpret when pooling the two entity types. Generally, we find large and significant increases in taxes and licenses deductions following the passage of PTET laws. Specifically, we find that PTETs increase S corporation taxes and licenses deductions by 19.8 percent (0.181 log units) and increase partnership taxes and licenses deductions by 40.5 percent (0.340 log units). Further, event study coefficients in the years before PTET passage are statistically indistinguishable from zero, lending credibility to our parallel trends assumption.

In Appendix Tables B.3 and **??**, we present several robustness checks and alternative specifications. Panel A of Table B.3 presents our main regression specification results, but adds results for sole proprietorship businesses, which are not able to take advantage of PTETs. As desired, we estimate a null treatment effect of PTETs on taxes and licenses deductions for sole proprietorships. In Panel B, we show the difference-in-differences version of the research design, where we do not account for C corporation growth. Here, we still estimate large and significant effects of the PTET laws for passthrough businesses (though smaller than in our main specification), but we also estimate significant negative effects for both sole proprietorships and C corporations, which are non-passthrough entity types. While this could be interpreted as resulting from business entity type switching, we think this is not likely given the discussion above. Instead, we view these negative coefficients for non-passthrough businesses as indicative of tax deduction changes unrelated to PTET regimes, further motivating our triple difference specification. Next, in Panel C, we show that our results are

statistically indistinguishable from a specification that includes the states with PTETs that operate through an income exclusion. Panel D shows that results are nearly identical when we do not use the stacked version of the regression. Finally, Panel E shows that results are nearly identical when we construct the triple difference specification with respect to sole proprietorships instead of C corporations.

Separately, in Table **??**, we show results from two alternative outcome specifications: first, the log of net business income, and second, taxes and licenses deductions as a share of total income. Given that PTETs operate through increased tax deductions at the expense of net business income, we expected to see negative effects of PTETs on the log of net income; however, these results were uninformative due to quite large standard errors. When measuring deductions as a share of total income we find results that are qualitatively similar to our primary results.

5 Fiscal Costs

In this second portion of our analysis, we estimate the federal fiscal cost of PTET deductions. Specifically, we first translate the causal estimates from the prior section into an estimate of aggregate federal business income deductions related to PTET payments. Then, we compute the amount of federal revenue that could be raised by shifting passthrough entity-level deductions back to the owners of the passthrough businesses, where SALT deductions are subject to the $10,000 limitation. We opt to estimate these costs for 2024, since it is the most recent completed tax year and includes the most up-to-date set of states with PTET regimes. However, this requires us to tether our estimate to forecasts of passthrough business income rather than values computed directly from tax data. In contrast, were we to estimate the fiscal cost using the most recent year for which we have complete tax data (2022), we would omit costs associated with the many states that have enacted PTETs since then.

In Table 3, we walk through the conversion our causal estimates into an estimate of aggregate PTET deductions claimed. This conversion occurs in three steps: first, we estimate the aggregate size of taxes and licenses deductions in PTET states, shown in Panel A; second, we use our estimated treatment effects from Section 4 to back out the implied value of taxes and licenses deductions attributable to PTET laws, shown in Panel B; and third, we allocate a portion of PTET deductions to passthrough business owners that are individual taxpayers

(as opposed to other businesses), shown in Panel C. Here, all dollar figures are expressed as billions of nominal 2024 dollars, and we show the computations separately for S corporations and partnerships.

To estimate aggregate taxes and licenses deductions in PTET states (Table 3, Panel A), we use our SOI samples to estimate these deductions as a share net ordinary business income (within PTET states) and extrapolate to 2024 using passthrough income growth forecasts from the Congressional Budget Office (CBO).[28] Line 1 is the share of passthrough income in states with PTET laws, approximated by matching the set of PTET laws in effect in 2024 to the 2022 SOI samples. Line 2 is taxes and licenses deductions as a share of net ordinary business income, based on the average share observed in the 2021 and 2022 SOI samples among states with PTET laws. Line 3 is total passthrough net ordinary business income, estimated using the 2022 SOI samples and extrapolated to 2024 using growth in passthrough income as forecasted by CBO. We multiply income by the two share parameters to obtain an estimate of the aggregate taxes and licenses deductions taken in PTET states.

We next apply our treatment effect estimates to estimate aggregate PTET deductions (Table 3, Panel B) . We do this by multiplying our estimates of aggregate taxes and licenses deductions from Panel A by $1 - \frac{1}{1+\tau}$, where τ is the estimated treatment effect expressed as a percent increase (divided by 100). This calculation returns an estimated $30.9 billion in PTET deductions for S corporations and an estimated $39.0 billion of deductions for partnerships, for a combined total of $69.8 billion.

Finally, we allocate partnership deductions to owners that are individual taxpayers (Table 3, Panel C). While S corporations are almost entirely owned by individual taxpayers, a majority of partnership income flows to non-individual entities, including C corporations, other partnerships, and foreign entities (Love, 2021).[29] Generally, only individual taxpayers face the SALT cap. Hence, in order to estimate fiscal costs of PTET laws, it is important to first estimate the share of partnership PTETs that benefit individual taxpayers. While we cannot estimate this share using our data (since we do not observe state PTET benefits), we believe it is likely to be quite high for two reasons. First, many states do not allow non-individual taxpayers to use PTET credits and instead levy the PTET only on the share of

[28]PTETs are generally applied to income broader than just ordinary business income, but we tether our extrapolation to ordinary business income since the measure better aligns with what CBO forecasts.

[29]S corporations may also be owned by estates, certain trusts, certain tax-exempt entities, and employee stock ownership plans. However, estimates from SOI data suggest nearly 95 percent of S corporation income flowed to individuals in 2022. In contrast, we estimate that 39.8 percent of partnership income flows to individuals. In computing this share, we disregard partnership income flowing to other partnerships, as we assume this income is eventually allocated to individuals, corporations, and/or tax-exempt entities.

partnership income that is allocated to individuals.[30] Second, even if non-individual partners are eligible to claim a state PTET credit, there is no federal tax benefit in the arrangement, which we believe will deter some of these partnerships from making a PTET election.[31] So, while only 40 percent of partnership income ultimately flows to individuals, we make an ad hoc assumption that 85 percent of partnership PTET deductions benefit individuals (the entire individual income share plus 75 percent of the non-individual share), though we explore robustness to this choice below. Hence, we estimate that $64.0 billion in SALT deductions were diverted from individual returns to business returns as a result of PTET regimes in 2024 (the entire $30.9 billion from S corporations plus a pro-rated $33.1 billion from partnerships).

We next simulate the fiscal consequences of denying PTET deductions to entities and routing the deductions back to individual taxpayers. To do this, we use several individual income tax parameters derived from the JCT Individual Tax Model (ITM). The ITM is a microsimulation model that is primarily used by JCT staff for the purposes of revenue estimation work for Congress, though, as noted below, this exercise differs somewhat from a traditional JCT revenue estimate. The microsimulation is based on the 2019 SOI Individual sample, which is then reweighted and extrapolated through the 10-year budget window (currently through fiscal year 2035) as to match dozens of aggregate variables from CBO economic and population forecasts. The ITM also includes a comprehensive income tax calculator. Here, we use the ITM to compute the deductible share of passthrough income via the individual deduction for qualified business income (QBI), the deductible share of SALT payments, and the average effective marginal tax rate on passthrough income. For more information on the JCT ITM, see Joint Committee on Taxation (2023).

Under our no-PTET hypothetical, business income passed through to owners would increase by the value of the denied PTET deduction, which would translate into increased taxable income on individual returns. However, some of that increase in passthrough business income would be reduced by deductions available to individuals: first, due to the Section 199A deduction for qualified business income, which reduces taxable income, and, second, due to

[30]Several of the largest states for partnership activity operate their PTETs this way, including California, New York, and Massachusetts. Other states compute PTETs with respect to all partnership income and allow C corporations to claim a corresponding credit, including Illinois and New Jersey.

[31]C corporations, who account for the majority of non-individual partnership owners, should be tax-indifferent to a partnership's PTET election (so long as they are entitled to a PTET credit or exclusion), since they can deduct SALT without limitation on Form 1120. However, foreign and tax-exempt entities may be left disadvantaged by a PTET election. Partnerships could deal with this uneven benefit by updating their partnership agreements to reallocate income to tax-exempt parties or even by creating an entirely new partnership to pool income generated by the individual owners.

taxpayers who have not yet exhausted SALT deductions allowed under the cap. Using the JCT ITM, we estimate that, in 2024, 8.7 percent of marginal passthrough income could be deducted as QBI and 1.3 percent of additional SALT liability could be deducted via itemizing deductions, where those shares represent an average among all individuals who report positive passthrough income, weighted by the amount of their passthrough income.[32]

After applying the deductible share parameters, we calculate that taxable passthrough income increases by $58.4 billion and that just $0.9 billion of additional PTET-related SALT could be deducted via itemizing. In other words, nearly all PTET deductions would be denied were they treated as individual SALT, but the effect on taxable income is offset somewhat due to the QBI deduction. Using the ITM, we estimate that in 2024, $217.5 billion in SALT was disallowed as a deduction for individual taxpayers who elected to itemize; individual-benefiting PTETs amount to 29 percent of this total. Framed differently, this implies that disallowed SALT deductions generated by the TCJA SALT cap are 22 percent lower due to PTET regimes (that is, using actual disallowed SALT plus would-be disallowed PTETs as the denominator). Finally, we use the JCT ITM to compute an income-weighted average MTR on passthrough income of 31.5 percent. Multiplying this rate by the implied increase in taxable income ($57.4 billion), we compute a fiscal cost of $18.1 billion. If we alternatively assume that partnership PTETs are borne by individual taxpayers at a rate exactly equal to individual ownership shares (40 percent), this cost figure drops to $13.1 billion. If we assume partnership PTETs are borne entirely by individual taxpayers, the cost figure increases to $19.7 billion.

We conclude this exercise by estimating how this fiscal cost is allocated across the income distribution. We distribute the fiscal cost of the SALT cap workarounds proportionately to individual passthrough income among taxpayers filing in states with an income tax and whose SALT liability exceeds $10,000.[33] This is reasonable because PTET taxes are generally levied on passthrough businesses as a fixed percentage of passthrough income. To add context to this exercise, we also use the JCT ITM to distribute the net fiscal effects of the SALT

[32]We estimate these shares by first adding $100 dollars in passthrough income to individuals and computing the dollar value change in QBI deductions, and then by adding $100 in SALT and computing the dollar value change in taxable income. For the latter computation, we find that taxable income decreases by $2.6, on average. However, because returning PTETs back to individuals almost certainly increase SALT by more than $100, we divide this $2.6 figure by 2 to represent the fact that a substantial increase in SALT will send many of these taxpayers to the SALT cap. This decision is virtually immaterial for the final cost estimate.

[33]Using filers in states with an income tax is only an approximation of filers who may benefit from a PTET because not all these states have a PTET regime and because non-residents could benefit from a PTET regime if they have a business interest in a PTET state and file a non-resident income tax return. Our results are essentially unchanged when we distribute proportionately to all individual passthrough income.

cap (both the TCJA and OBBBA caps) relative to an unlimited SALT deduction. These SALT cap effects can be thought of as the marginal revenue raised by the caps when fixing individual tax situations (i.e., it assumes no behavioral response to an unlimited SALT cap other than tax form optimization, namely via the decision to itemize). When distributing fiscal effects, we allocate tax returns into percentile bins based on a comprehensive economic income concept. This income concept corresponds to the definition used in JCT distribution tables and is equal to AGI plus a suite of non-taxed income and compensation items.[34] To rank tax returns and compute percentiles, we adjust economic income for the size of a tax unit by dividing it by the square root of the number of individuals in the tax unit. This adjustment is standard in the study of the distribution of household income.

Our distribution analysis, presented in Table 4, shows that SALT cap workaround fiscal effects are concentrated at the top of the income distribution, even more so than the effects of the TCJA SALT cap. Specifically, we find that about 80 percent of the PTET fiscal cost is generated by the top 1 percent of the income distribution, but that only about 55 percent of the fiscal benefit from the SALT cap (a revenue raiser) is generated by the top 1 percent. Further, we find that 94 percent of PTET fiscal costs are generated by the top 5 percent. Note that our estimate of the fiscal cost of the workarounds is computed relative to the TCJA SALT cap, which differs from the OBBBA cap. While the new OBBBA SALT cap will affect the fiscal cost estimate, the interaction will be small, since the OBBBA-instituted $40,000 SALT cap phases down to $10,000 for taxpayers with modified adjusted gross income between $500,000 and $600,000. Indeed, we find that the OBBBA cap did not meaningfully affect the SALT cap fiscal benefit generated from the top 1 percent of the income distribution, relative to the TCJA cap. For this reason, the fiscal costs of the PTET workarounds are likely to remain largely intact despite the modified SALT cap.

This exercise of computing SALT cap workaround fiscal costs differs from a JCT revenue estimate in a few crucial ways. First, it is retrospective and corresponds to a single tax year, while JCT revenue estimates are prospective, correspond to fiscal year costs, and are presented over a 10-year budget window. More importantly, however, this cost estimate is not tethered to a legislative proposal and does not incorporate any taxpayer or state government behavioral responses to possible legislative changes to the permissibility of SALT cap workarounds. For example, a legislative proposal narrowly targeting passthrough entity

[34]Specifically, economic income is defined as adjusted gross income plus 1) tax-exempt interest, 2) employer contributions for health plans and life insurance, 3) employer share of FICA tax, 4) workers' compensation, 5) nontaxable Social Security benefits, 6) insurance value of Medicare benefits, 7) alternative minimum tax preference items, 8) individual share of business taxes, and 9) excluded income of U.S. citizens living abroad.

income tax workarounds may be met by state governments enacting new workarounds via sales taxes, property taxes, or any other payment that could generate a corresponding state individual tax benefit. A JCT revenue estimate would consider these (and other) behavioral responses.

6 Conclusion

In this paper, we document the role of state PTET regimes in diverting individual state income tax deductions to federal passthrough business tax returns, which we estimate translated to about $18 billion in foregone federal tax receipts in 2024. Framed differently, PTETs have negated more than one-fifth of would-be disallowed SALT deductions had these laws not existed. The prevalence of PTETs highlights the important role of state tax policy in shaping the effectiveness of federal tax policy. While tax-related behavioral responses by individuals and business are widely studied and frequently incorporated into tax policy analysis, responses by state and local governments are less often considered. The proliferation of PTET regimes should not be taken as a "one-off" response by state governments; in Section 2, we document a number of attempts by states to facilitate SALT cap workarounds. Hence, state government reactions should be considered by policy analysts whenever federal tax policy modifies the deductibility of state and local taxes, and possibly even more broadly.

During the past year, there has been notable legislative interest in limiting SALT cap workarounds, with related provisions included in the version of OBBBA passed by the House on May 22 and also in a draft released by the Senate Finance Committee on June 16.[35] Both the House and Senate proposals broadly denied a PTET deduction to partnerships and S corporations and required them to separately state these deductions to individual partners and shareholders for their inclusion toward the SALT cap. However, neither proposal entirely eliminated the PTET benefit: the House version allowed certain qualifying passthrough businesses such as manufacturers and retailers an unlimited PTET deduction, whereas the Senate Finance Committee version subjected all separately stated PTETs to a more generous modified SALT cap. While some commentators criticized these early OBBBA drafts for allowing some passthrough businesses to continue deducting PTETs,[36] others argued that

[35]For the House provision, see Section 112018 of the bill as engrossed in the House, available at `https://www.congress.gov/bill/119th-congress/house-bill/1/text/eh?s=4&r=12`. For the Senate Finance Committee provision, see Section 70601 of the June 16, 2025 Committee print, available at `https://www.finance.senate.gov/imo/media/doc/finance_committee_legislative_text_title_vii.pdf`.

[36]The Editorial Board, "Closing the SALT Business Loophole," Wall Street Journal, June 24, 2025.

any limitation on SALT cap workarounds violated tax neutrality and that the proposals imposed unnecessary administrative burdens on taxpayers.[37] Ultimately, the Senate opted to remove proposed limits on SALT cap workarounds in the final version of OBBBA, which was then passed without amendments by the House. However, these draft provisions may very well serve as a basis for future legislative proposals.

[37]Parillo, "SALT Cap Workaround Limit Raises Fairness Concerns," Tax Notes Federal, June 9, 2025, p. 1967.

References

Baker, Andrew C., David F. Larcker, and Charles C.Y. Wang, 2022, "How much should we trust staggered difference-in-differences estimates?" *Journal of Financial Economics* 144 (2), 370–395.

Bloomberg Tax Research, 2025, "Custom State Tax Tables," Accessed: September 9, 2025.

Boutwell, George S., 1864, *Manual of the Direct and Excise Tax System in the United States: Including the Forms & Regulations Established by the Commissioner of Internal Revenue, the Decisions & Rulings of the Commissioner*, 4th ed., Little, Brown, and Company.

Butters, R. Andrew, Daniel W. Sacks, and Boyoung Seo, 2022, "How Do National Firms Respond to Local Cost Shocks?" *American Economic Review* 112 (5), 1737–1732.

Cengiz, Doruk, Arindrajit Dube, Attila Lindner, and Ben Zipperer, 2019, "The Effect of Minimum Wages on Low-Wage Jobs," *The Quarterly Journal of Economics* 134 (3), 1405–1454.

Chodorow-Reich, Gabriel, Matthew Smith, Owen Zidar, and Eric Zwick, 2024, "Tax Policy and Investment in a Global Economy," NBER Working Paper 32180.

Deshpande, Manasi, and Yue Li, 2019, "Who is Screened Out? Application Costs and the Targeting of Disability Programs," *American Economic Journal: Economic Policy* 11 (4), 213–248.

Feenberg, Daniel R., and Harvey S. Rosen, 1986, "The Interaction of State and Federal Tax Systems: The Impact of State and Local Tax Deductibility," *American Economic Review* 76 (2), 126–131.

Feldstein, Martin S., and Gilbert E. Metcalf, 1987, "The Effect of Federal Tax Deductibility on State and Local Taxes and Spending," *Journal of Political Economy* 95 (4), 710–736.

Gale, William G., and Claire Haldeman, 2021, "The Tax Cuts and Jobs Act: Searching for Supply-Side Effects," *National Tax Journal* 74, 895–914.

Goodman, Lucas, Katherine Lim, Bruce Sacerdote, and Andrew Whitten, 2025, "How do business owners respond to a tax cut? Examining the 199A deduction for pass-through firms," *Journal of Public Economics* 242, 105293.

Goodman-Bacon, Andrew, 2021, "Difference-in-differences with variation in treatment timing," *Journal of Econometrics* 225 (2), 254–277.

Horton, Emily, 2023, "Are tax benefits priced into home values? Evidence from the Tax Cuts and Jobs Act," Working Paper.

Joint Committee on Taxation, 2023, "Estimating Changes in the Federal Individual Income Tax: A Description of the Individual Tax Model for 2023," JCX-48-23.

Joint Committee on Taxation, 2024, "Estimation of Federal Tax Expenditures for Fiscal Years 2024-2028," JCX-48-24.

Kennedy, Patrick, Christine Dobridge, Paul Landefeld, and Jacob Mortenson, 2024, "The Efficiency-Equity Tradeoff of the Corporate Income Tax: Evidence from the Tax Cuts and Jobs Act," Working Paper.

Kessler, Lawrence, and Donald Bruce, 2024, "A SALT on real estate? housing market and migration responses to the limit on the state and local tax deduction," *Contemporary Economic Policy* 42, 683–704.

Love, Michael, 2021, "Where in the World Does Partnership Income Go? Evidence of a Growing Use of Tax Havens," Working Paper.

New York State Division of the Budget, 2025, "Fiscal Year 2026 Annual Report on New York State Tax Expenditures," Department of Taxation and Finance.

Rueben, Kim S., 2021, "State Pass-Through Entity Taxes Let Some Residents Avoid the SALT Cap at No Cost to The States," *Tax Policy Center: TAXVOX* .

Smith, Matthew, Danny Yagan, Owen Zidar, and Eric Zwick, 2019, "Capitalists in the Twenty-First Century," *Quarterly Journal of Economics* 134 (4), 1675–1745.

Smith, Matthew, Danny Yagan, Owen Zidar, and Eric Zwick, 2022, "The Rise of Pass-Throughs and the Decline of the Labor Share," *American Economic Review: Insights* 4, 323–340.

Statistics of Income, 2019, "Individual Income Tax Returns Line Item Estimates, 2017," Internal Revenue Service, Publication 4801 (Rev. 9–2019).

Statistics of Income, 2020, "Individual Income Tax Returns Line Item Estimates, 2018," Internal Revenue Service, Publication 4801 (Rev. 9–2020).

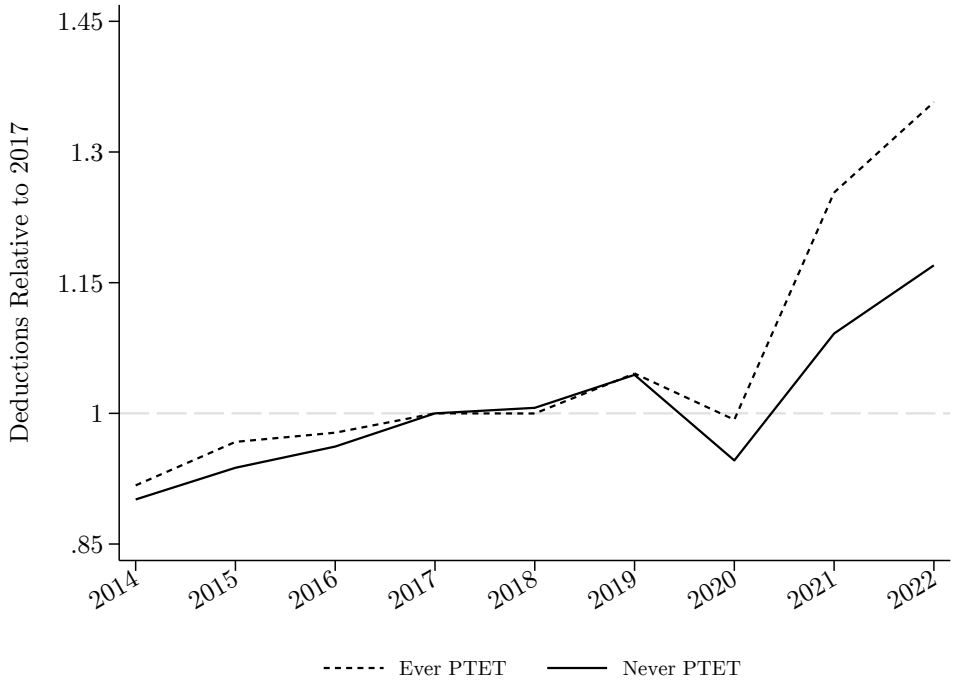
Statistics of Income, 2025a, "Individual Income Tax Returns Complete Report, 2022," Internal Revenue Service, Publication 1304 (Rev. 1–2025).

Statistics of Income, 2025b, "SOI Tax Stats - Sampling methodology and data limitations by study," `https://www.irs.gov/statistics/soi-tax-stats-sampling-methodology-and-data-limitations-by-study`.

Wing, Coady, Seth M. Freedman, and Alex Hollingsworth, 2024, "Stacked Difference-in-Differences," NBER Working Paper 32054.

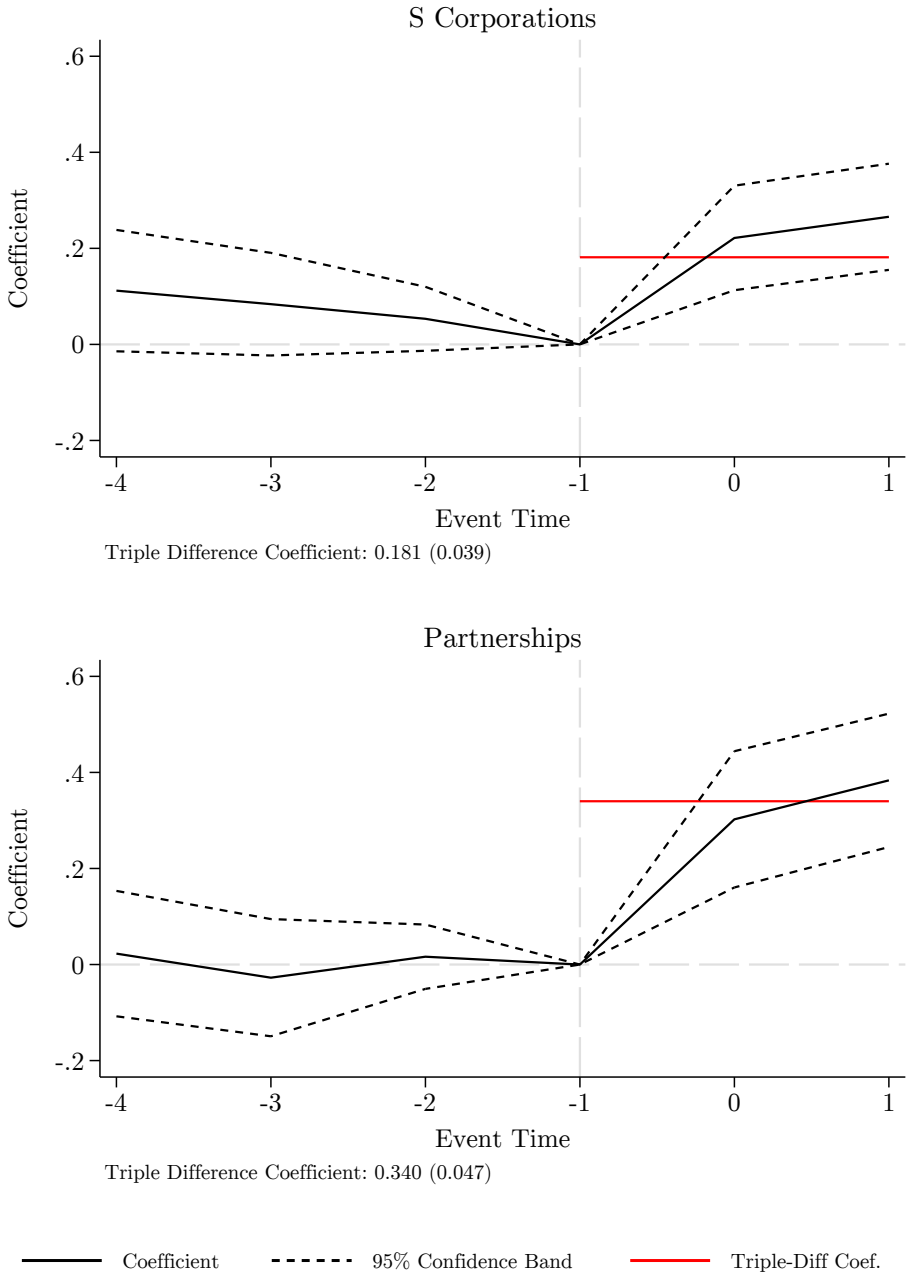
Figures

Figure 1: Passthrough Entity Taxes & Licenses Deduction, Relative to 2017



"Ever PTET" states are those that passed a PTET law by 2022 (the last year of our SOI data), and "Never PTET" are states that did not pass a PTET law by 2022. Deductions are inflation adjusted and then scaled to 2017 values.

Figure 2: Effect of PTET Laws on Log Taxes & Licenses Paid Deduction



Coefficients from stacked version of Equation 2, where the top panel includes S corporations and C corporations and the bottom panel includes partnerships and C corporations. Triple difference coefficient line in red.

Tables

Table 1: Overview of PTET Laws as of 2025

State	PTET Enacted	PTET Rate	Top MTR	Enacted Year	Method
AK	X	-	0.00%	-	-
AL	✓	5.00%	5.00%	2021	Credit
AR	✓	3.90%	3.90%	2021	Exclusion
AZ	✓	2.50%	4.50%	2021	Credit
CA	✓	9.30%	13.30%	2021	Credit
CO	✓	4.40%	4.40%	2021	Credit
CT	✓	6.99%	6.99%	2018	Credit
DC	X	-	10.75%	-	-
DE	X	-	6.60%	-	-
FL	X	-	0.00%	-	-
GA	✓	5.19%	5.19%	2021	Exclusion
HI	✓	9.00%	11.00%	2023	Credit
IA	✓	3.80%	3.80%	2023	Credit
ID	✓	5.30%	5.30%	2021	Credit
IL	✓	4.95%	4.95%	2021	Credit
IN	✓	3.00%	3.00%	2023	Credit
KS	✓	5.58%	5.58%	2022	Credit
KY	✓	4.00%	4.00%	2023	Credit
LA	✓	3.00%	3.00%	2019	Exclusion
MA	✓	5.00%	9.00%	2021	Credit
MD	✓	8.75%	6.50%	2020	Credit
ME	X	-	7.15%	-	-
MI	✓	4.25%	4.25%	2021	Credit
MN	✓	9.85%	9.85%	2021	Credit
MO	✓	4.80%	4.70%	2022	Credit
MS^{+}	✓	5.00%	4.40%	2022	Credit
MT	✓	5.90%	5.90%	2023	Credit
NC	✓	4.25%	4.25%	2021	Exclusion
ND	X	-	2.50%	-	-
NE	✓	5.20%	5.20%	2023	Credit

Continued on next page

Table 1: Overview of PTET Laws as of 2025

State	PTET Enacted	PTET Rate	Top MTR	Enacted Year	Method
NH*	X	-	0.00%	-	-
NJ+	✓	10.90%	10.75%	2020	Credit
NM	✓	5.90%	5.90%	2022	Credit
NV	X	-	0.00%	-	-
NY+	✓	10.90%	10.90%	2021	Credit
OH	✓	3.00%	3.13%	2022	Credit
OK	✓	4.75%	4.75%	2019	Exclusion
OR+	✓	9.90%	9.90%	2021	Credit
PA	X	-	3.07%	-	-
RI	✓	5.99%	5.99%	2019	Credit
SC	✓	3.00%	6.20%	2021	Exclusion
SD	X	-	0.00%	-	-
TN	X	-	0.00%	-	-
TX	X	-	0.00%	-	-
UT	✓	4.50%	4.50%	2022	Credit
VA	✓	5.75%	5.75%	2022	Credit
VT	X	-	8.75%	-	-
WA**	X	-	0.00%	-	-
WI	✓	7.90%	7.65%	2018	Exclusion
WV	✓	4.82%	4.82%	2023	Credit
WY	X	-	0.00%	-	-

Note: Table shows 2025 PTET tax rates and top marginal tax rates for individuals, as applicable to passthrough income. Local income taxes are not included, even if collected on state returns, as is the case in Iowa, Indiana, Maryland, and New York. Only Maryland's PTET rate is set under consideration of local income tax rates.

{+} PTET schedule is graduated; only top rate shown.

{*} State only taxes interest and dividend income.

{**} State only taxes capital gains income.

Source: Bloomberg Tax Research (2025).

Table 2: Example of PTET Benefit Under a Hypothetical Credit Regime

	Basic Case	PTET Regime
A. Passthrough Business Income		
Total Passthrough Income	1,000,000	1,000,000
State PTET Rate	0%	5%
Passthrough PTET Tax	0	50,000
Distributed Passthrough Income	1,000,000	950,000
B. Shareholder State Income Tax		
Allocated State Passthrough Income	1,000,000	1,000,000
State Taxable Income	1,000,000	1,000,000
State Individual ATR	5%	5%
State Tax, Before Credits	50,000	50,000
State PTET Credit	0	50,000
State Tax Liability	50,000	0
Passthrough Income After State Tax	950,000	950,000
C. Shareholder Federal Income Tax		
Allocated Federal Passthrough Income	1,000,000	950,000
Total Individual SALT	50,000	0
Allowable SALT Deduction	10,000	0
Federal Taxable Income	990,000	950,000
Federal Individual ATR	35%	35%
Federal Tax Liability	346,500	332,500
Passthrough Income After All Tax	603,500	617,500

Table 3: Estimates of Aggregate and Individual PTET Deductions (2024)

	S Corps	Partnerships	Total
A. Forecast of PTET State Taxes & Licenses Deductions			
Share of Income in PTET States	0.717	0.615	-
Taxes & Licenses Deductions as a Share of Income	0.313	0.219	-
Total Ordinary Business Income ($ Billions)	832.2	1,003.7	-
PTET State Taxes & Licenses Deductions ($ Billions)	186.8	135.2	321.9
B. Estimated Aggregate PTET Deductions			
PTET Treatment Effect (Pct. Increase / 100)	0.198	0.405	-
Aggregate PTET Deductions ($ Billions)	30.9	39.0	69.8
C. Estimated Individual PTET Deductions			
Partnership Individual Income Share	-	0.398	-
Additional Share (75% Selection Adjustment)	-	0.452	-
Individual PTET Deductions ($ Billions)	30.9	33.1	64.0

Notes: Panel A and C values derived from 2019-2022 SOI samples, extrapolated to 2024. Taxes & Licenses deductions as a share of income is computed only with respect to PTET states. Panel B estimates converted from log effects estimated via Equation 1.

Table 4: Distribution of Net Fiscal Effects, Tax Year 2024

	TCJA SALT Cap		OBBBA SALT Cap		SALT Cap Workarounds	
Income Quantile	Billions	Percent	Billions	Percent	Billions	Percent
First Quantile (Less than P_{20})	0.0	0.0%	0.0	0.0%	0.0	0.0%
Second Quantile (P_{20} to P_{40})	0.0	0.0%	0.0	0.0%	0.0	0.1%
Third Quantile (P_{40} to P_{60})	0.2	0.2%	0.0	0.0%	0.0	0.3%
Fourth Quantile (P_{60} to P_{80})	1.5	1.9%	0.0	0.0%	-0.1	0.8%
80th–90th percentile	4.2	5.3%	0.0	0.1%	-0.3	1.6%
90th–95th percentile	7.3	9.1%	0.2	0.3%	-0.5	3.0%
95th–99th percentile	23.1	28.8%	8.5	16.6%	-2.5	14.1%
99th–99.9^{th} percentile	25.5	31.8%	24.6	47.6%	-6.9	38.3%
Highest 0.1 Percentile	18.3	22.8%	18.3	35.4%	-7.6	41.9%
Total, All Taxpayers	80.1	100%	51.6	100%	-18.1	100%

Notes: SALT cap fiscal effects are computed relative to a counterfactual of an unlimited SALT cap. The OBBBA SALT cap simulation applies 2025 SALT cap parameters to tax year 2024. SALT cap workarounds costs are computed relative to the TCJA SALT cap. Fiscal costs do not include effects related to taxpayer behavior, other than tax form behavior such as the decision to itemize. See text for definition of income concept.

A Data Appendix

We construct our main sample of business tax returns by iterating through each SOI Corporate and Partnership sample from 2014 to 2022. All businesses in the Corporate and Partnership samples are included in our main analysis.

As a robustness check, we use the Individual Sample to examine sole proprietorships. The Individual Sample is comprised of individual taxpayers. Some taxpayers, in addition to their individual tax return, may file a Schedule C to report profit or loss from a non-farm business they own. We narrow the Individual Sample to only include taxpayers filing a Schedule C. Tax units may file multiple Schedule Cs if they own multiple businesses. Since each observation in the Individual Sample is meant to represent one tax unit, we alter the configuration of the dataset such that each row represents a business. For example, if a tax unit filed two Schedule Cs, this tax unit would appear twice in our dataset – once for each Schedule C.

These businesses file income tax returns through a variety of IRS forms. Table A.1 lists the specific line items we retrieve from each of the four most common business income tax forms in our extended analysis.

Table A.1: Line Items Retrieved from Each Business Tax Return

Line Item	Form 1120	Form 1120-S	Form 1065	Form 1040, Schedule C
Gross Receipts	Line 1a	Line 1a	Line 1a	Line 1
Total Income	Line 10	Line 6	Line 8	Line 7
Taxes & Licenses	Line 17	Line 12	Line 14	Line 23
Total Deductions	Line 27	Line 21	Line 22	Line 28
Taxable Income	Line 30	Line 22	Line 23	Line 31

Note: Lines correspond to Tax Year 2022 Forms.

B Appendix Tables

Table B.1: PTET Law Enacted Dates and Effective Years

State	Date of Enactment	Effective Year	Method
AK	-	-	-
AL	2/12/2021	2021	Credit
AR	3/15/2021	2022	Exclusion
AZ	7/9/2021	2022	Credit
CA	7/16/2021	2021	Credit
CO	6/23/2021	2018	Credit
CT	5/31/2018	2018	Credit
DC	-	-	-
DE	-	-	-
FL	-	-	-
GA	5/4/2021	2022	Exclusion
HI	6/1/2023	2023	Credit
IA	5/11/2023	2022	Credit
ID	4/15/2021	2021	Credit
IL	8/27/2021	2021	Credit
IN	2/22/2023	2022	Credit
KS	4/14/2022	2022	Credit
KY	3/31/2023	2022	Credit
LA	6/22/2019	2019	Exclusion
MA	9/30/2021	2021	Credit
MD	5/8/2020	2020	Credit
ME	-	-	-
MI	12/21/2021	2021	Credit
MN	7/1/2021	2021	Credit
MO	6/30/2022	2022	Credit
MS	4/14/2022	2022	Credit
MT	5/19/2023	2023	Credit
NC	11/18/2021	2022	Exclusion
ND	-	-	-

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Table B.1: PTET Law Enacted Dates and Effective Years

State	Date of Enactment	Effective Year	Method
NE	5/31/2023	2018	Credit
NH	-	-	-
NJ	1/13/2020	2020	Credit
NM	3/8/2022	2022	Credit
NV	-	-	-
NY	4/19/2021	2021	Credit
OH	6/14/2022	2022	Credit
OK	4/29/2019	2019	Exclusion
OR	7/19/2021	2022	Credit
PA	-	-	-
RI	7/5/2019	2019	Credit
SC	5/17/2021	2021	Exclusion
SD	-	-	-
TN	-	-	-
TX	-	-	-
UT	3/23/2022	2022	Credit
VA	4/11/2022	2021	Credit
VT	-	-	-
WA	-	-	-
WI	12/14/2018	2018	Exclusion
WV	6/8/2023	2022	Credit
WY	-	-	-

Table B.2: Alternative Regression Specifications

	S Corps	Partnerships	Sole Props	C Corps
A. Main Regression Results				
Treat × Post	0.181***	0.340***	-0.021	-
	(0.039)	(0.064)	(0.069)	-
B. Add Exclusion States				
Treat × Post	0.173***	0.317***	-0.021	-
	(0.038)	(0.048)	(0.065)	-
C. Diff-in-Diff				
Treat × Post	0.104***	0.226***	-0.0954**	-0.102**
	(0.020)	(0.041)	(0.0439)	(0.0455)
D. No Stacking				
Treat × Post	0.178***	0.313***	-0.005	-
	(0.038)	(0.065)	(0.073)	-
E. Triple Diff w.r.t. Sole Props				
Treat × Post	0.192***	0.359***	-	0.021
	(0.046)	(0.058)	-	(0.069)

Notes: Panel A shows the main regression specification of a triple difference regression on a "stacked" panel. Panel B adds back in treated states with an "income exclusion" PTET law. Panel C shows a naive difference-in-differences regression that does not control for C corporation trends. Panel D shows a version of the regression on a standard panel (i.e., not stacked).
Significance Levels: * 10 percent; ** 5 percent; *** 1 percent.

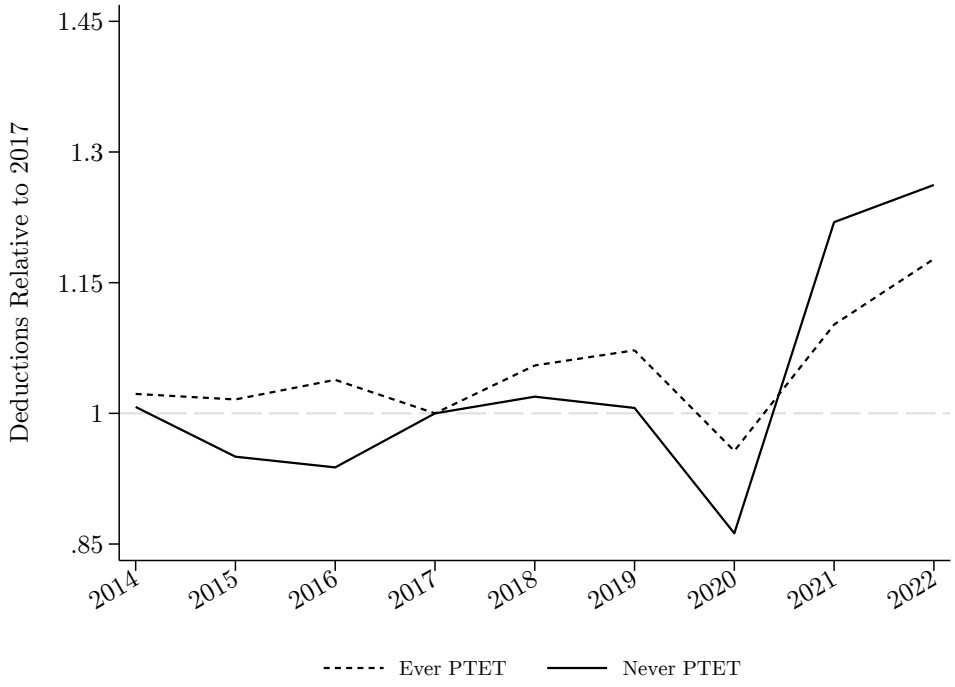
Table B.3: Alternative Regression Outcomes

	S Corps	Partnerships	Sole Props
A. Main Regression Results			
Treat × Post	0.181***	0.340***	-0.021
	(0.039)	(0.064)	(0.069)
B. Log Net Income			
Treat × Post	0.013	-0.165	0.010
	(0.101)	(0.240)	(0.098)
B. Taxes & Licenses as a Share of Total Income			
Treat × Post	0.007***	0.010***	0.001
	(0.002)	(0.002)	(0.002)

Notes: Panel A shows the main regression results, using log Taxes & Licenses deductions as the outcome. Panel B shows results using log of net business income as an outcome. Panel C shows results using the ratio of Taxes & Licenses deductions to total income as the outcome.

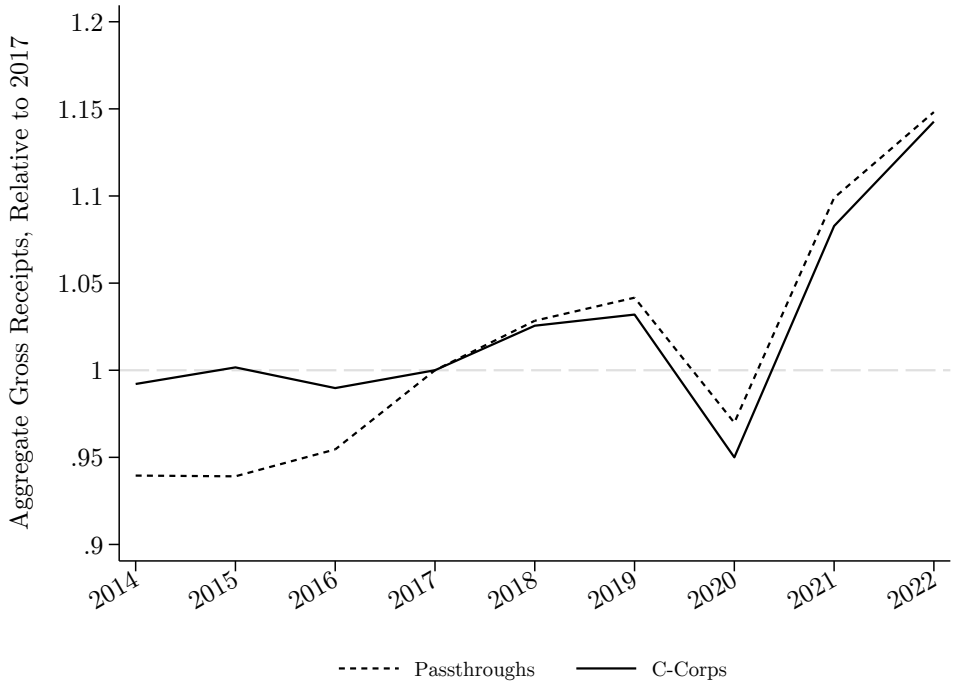
C Appendix Figures

Figure C.1: C Corporation Taxes & Licenses Deduction, Relative to 2017



"Ever PTET" states are those that passed a PTET law by 2022 (the last year of our SOI data), and "Never PTET" are states that did not pass a PTET law by 2022. Deductions are inflation adjusted and then scaled to 2017 values.

Figure C.2: PTET State Gross Receipts Trends by Entity Type, Relative to 2017



Sample is restricted to states that passed a PTET law by 2022. Gross receipts are inflation adjusted and then scaled to 2017 values.